DEVON THIELE

The Thiele Machine

What survives.

A BOOK IN TWO DIRECTIONS

Read from either end.

From this end, the words are from the released book. From the other, they are from four earlier versions of the same passages. Turn the volume over to enter the other reading.

Both readings use one setting of type. A word that belongs to only one version keeps its place in the other as an absence. Nothing moves in to cover it.

The earlier wording is historical material, not the book's current claim. The small number at the foot takes you to the other face of the passage.

What I am is curious, past the point of good sense.

I pick a thing up and cannot set it down
until I have turned it over enough times
to see how it holds together, and this
time it did not let go.

Then I hit a wall, and it is the wall this whole document is about. Z3 could settle the small questions and fell over on the exact ones I needed answered, which left me staring down an infinite hallway of guess-and-test with no proof waiting at the end of it, ever. One night, not even working, I was watching a video about Flatland, the old story about a creature living in two dimensions who cannot so much as picture a third. And it landed on me sideways, all at once, the way the real ones always do. I said it out loud: I get it now. There is another axis. Classical computation only ever sees what it can read and write off its own surface, and there is a whole direction it is blind to. It

isn't broken. It was just never built with an eye that points that way. That's how it landed that night, and it didn't survive checking in that form: the direction turned out to be one an ordinary computer can reach, and what the usual rules leave out is the price (Section 3.6 says where the axis ended up). The two computers on the first pages of this book are my attempt to show you that blindness with something you can hold. I opened a new repo that same night and called the thing in it the Thiele Machine. The name began as a working title and became the name of the model. It carries the larger claim plainly: the machine treats structural state as part of computation.

It isn't more computing power. Anything a Thiele machine computes, the machine under its record computes without it. The book proves that over any machine at all, a base-driven record is a switch the machine's own events flip once and nothing flips back; it watches the computing and adds nothing to it.

That's a sentence about the record. The checker belongs to the machine under it, and the definition of Thiele-complete doesn't ask the checker to be a program. Hand it a checker that knows which programs

halt, and whether a CHECK of “this program halts” traps tells you whether it does, so that machine decides halting (Section 16 has the details). With a checker that is a program, as every checker in this book is, nothing in the machine goes past what a program can do. The record makes nothing faster either: erase the flag and the counter, and every run takes the same steps to the same places. CHECK is another matter. It's a test at unit cost, and one CHECK of “A ≥ m” settles whether A ≥ m in one step, where the two

counters alone would count A down. It isn't quantum. It says nothing about which problems are hard.

The second says that even if reading were free, a fixed program couldn't keep them by sending each guest property to one of a fixed, finite list of host properties and checking it on the same register the guest's fact is about. That's all the theorem below rules out, and it's the pigeonhole.

Of all the events a machine could put on its meter, the ones its price list charges for and its counter adds up, certification is the one whose records get copied and relied on by parties who didn't produce them: checked, stored, and acted on across independent places that share no other state. Copying alone can't be the test. A replicated system copies everything it holds to every node, the gas used in a block header as faithfully as the word final, so rival events get copied too. Nobody who didn't produce

them acts on them. If that's true, the choice of certification stops being free. It's the one event the world around the machine keeps redundant copies of and leans on, and that's a reason to price it and leave the others unpriced. So the thesis has two halves, and the ways to kill it at the end aim at both: certification is that event, and where a deployed discipline puts a price on writing a record, the record it prices is one that's copied and relied on in that sense.

A word on what I actually do, because it runs under everything after it. I do not write code or math by hand. I do not pick constants. I do not grind through derivations. I have never written a line of code on paper or in a notepad in my life. What I do is see, and direct, and refuse. When I take hold of an idea it does not stay a point. It opens up. It becomes a shape I can blow up and turn over and take apart, lay across another shape to find where the two of them touch, run forward and then back to where it started, zoom into until it is the only thing in the world and pull back until it is one dot in a field of other dots, and the whole time it stays wired to everything I have ever connected it to, the right connections and the wrong ones both, all of them live at once. I am not reading that field neutrally, the way a search would. The thing doing the looking is me, the whole of me, my gut and my mood and my stubbornness picking what lights up and which way I move through it, and that is the part you cannot lift out without turning me back into a search engine. So I see where a thing has to fit, the shape it is obligated to have, usually well before I could write it down in a single symbol. Then the model gives me the symbols, the proof checker takes them or throws them back, and I refuse anything that will not compile or does not match the shape I am holding. That refusal is the one thing in this entire document that is mine and only mine: the demand that it be true, and the throwing-out of everything that wasn't.

So that's the Thiele Machine. An abstract model with five parts and one rule, an argument made with it that the usual account of a computation is a shadow and the axis it leaves out belongs in the step, and one small machine to check the argument against, with one fixed program, written in its instruction kinds plus one paid move and run on a host with more counters, that runs every machine of the model whose workings a program can present. The theorems are what they are, with their premises printed. The interpretation is mine, and I've tried to say at every turn which is which, because the first time through I was the only one checking, and I've met me, and that's not enough people. Now there's you.

The construction

Each pair of passages is aligned word by word. Shared words occupy the same positions. A removed word appears only in the earlier reading; an added word appears only in the current reading. Both reserve the same amount of paper in both directions.

The spaces are consequences of these actual edits. They do not measure the machine's structural cost, the author's labor, or a reader's understanding. A displayed history is not a proof that a claim is true.

01 The first shape

Change 39b6713de61c · parent 826ac7df66a7

02 What the record does

Change ce093c906465 · parent 3e8b497087d9

03 What a program cannot do

Change 5e1317334058 · parent 8d7f15d74486

04 What deserves a record

Change 8d79b65670b7 · parent 859c7aeae864

The object

Devon Thiele is the author. The prose excerpts are taken from the manuscript. The short reading directions and this colophon belong to this design study; they are not additions to the released monograph.

Section numbers are resolved against the released book. TeX emphasis becomes roman type and mathematical source notation becomes printed symbols. No claim is silently rewritten. The source bundle preserves both TeX passages, both readings, and every word’s position.

One PDF; ordinary opaque paper. The second half is intentionally inverted and sequenced from the other cover. Duplex printing uses the long edge, at actual size. No transparency, cutting, special ink, animation, or external device supplies part of the reading.

Reading proof, October 2026. CC-BY-SA-4.0. Fonts: URW Palladio and Nimbus Roman/Sans, embedded. This is an experiment in the book’s form, not the final design of the full monograph.

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This is what changed.

The words in this direction are earlier wording. They are printed here because the book changed them. Read from the other cover for the surviving passages.

The missing words are present at the other end. Their places are kept even on a page that would otherwise be empty. The revision has a volume.

All four comparisons come from the repository. They are excerpts, not a complete account of the book's history. Source locations meet in the middle.

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What it used to say.

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